Sectoral macroeconomic damage functions of weather extremes in Europe

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Data

Projection of weather extremes for 5 SSPs-RCPs for 1644 regions in 82 countries is available at this link.

Variable definitions

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Table 1: Climate indicators: definitions and construction.

Indicator	**Description**
TX	Mean daily maximum temperature (°C). Daily maxima are averaged to monthly values, then to an annual mean.
TN	Mean daily minimum temperature (°C). Daily minima are averaged to monthly values, then to an annual mean.
TM	Mean daily temperature (°C). Computed as the average of monthly TX and TN, then averaged to an annual mean.
TVAR	Intra-annual temperature variability (°C). Monthly standard deviation of daily mean temperature around the monthly mean, averaged across months.
RR	Total annual precipitation (mm). Sum of daily rainfall over the calendar year.
RX	Maximum 5-day cumulative precipitation (mm). The highest rolling 5-day rainfall total recorded during the year.
SNX	Maximum 3-day cumulative snowfall (mm). The highest rolling 3-day snowfall total recorded during the 6-month winter season.
WD	Wet days (days year^{-1}). Count of days with daily precipitation exceeding 1 mm.
PEXT	Extreme precipitation (mm year^{-1}). Annual sum of daily precipitation on days exceeding the climatological 99.5th percentile threshold, derived from daily 99th percentile values over the 1981–2010 baseline.
HW	Heatwave severity (°C·days year^{-1}). Cumulative sum of daily TX exceedances above the smoothed 90th-percentile threshold (1981–2010 baseline) during events of $\geq$5 consecutive days where TX also exceeds 30°C, summed over the 6-month summer season.
CW	Cold wave severity (°C·days year^{-1}). Analogous to HW but computed using TN during the 6-month winter season, with an absolute threshold of 0°C.
PET	Potential evapotranspiration (mm year^{-1}). Estimated using the Hargreaves formula from monthly TX, TN, and latitude, averaged to an annual value.
SPI	Standardised Precipitation Index (unitless). Annual sum of monthly SPI-6 values during drought events (defined as $\geq$3 consecutive months with SPI ≤ -1, continuing while SPI ≤ 0), excluding hyper-arid (AI ≤ 0.05) and frozen areas (mean monthly PET < 30.4 mm month^{-1}); reference period 1950–2010, Gamma distribution.
SPEI	Standardised Precipitation–Evapotranspiration Index (unitless). Analogous to SPI but incorporating the precipitation–PET balance; log-logistic distribution, same masking and drought event definition.

Usage

The coefficients were estimated with a variable selection process (Post-LASSO). Use the coefficients in a linear model as

$$\text{GDP per capita growth}_{it} = \text{Baseline growth}_{it} + \beta_1 \Delta_{t,t0} Extreme\ 1_{it} + \beta_2 \Delta_{t,t0} Extreme\ 2_{it} + \ldots$$

for country i and time t. We do not report standard errors as Post-LASSO regression standard errors are invalid for inference.

Damage coefficients - Europe

Table 3: Post-LASSO regression results for the growth rate of gross regional product per capita and its sectoral equivalents for Europe.

	(1) GRP	(2) Agriculture	(3) Manufacturing	(4) Services
PET	0.000215		0.000236	0.000192
RX	0.000179			0.000265
TVAR	-0.0423		-0.0396	-0.0559
SPEI	-0.0000660	-0.00290	-0.000915	-0.000427
CW	0.0000602	0.000183		-0.000000178
TN	-0.00904		-0.0201	-0.00918
HW		-0.00137		0.000106
RR		-0.000111		
N	9958	8811	8814	8806
Adjusted R2	0.640	0.408	0.624	0.603
Adjusted within-R2	0.0147	0.0196	0.0145	0.0199

Notes: The table presents the results of the Post-LASSO regression. The dependent variable is the growth rate of gross regional product or of its sectoral equivalent. The independent variables are weather extremes and have been selected through Adaptive LASSO. All regressions include region fixed effects, year fixed effects, and regional linear time trends. Note that standard errors for the Post-LASSO estimates are invalid for inference.

Damage coefficients - Global

Table 4: Post-LASSO regression results for the growth rate of gross regional product per capita and its sectoral equivalents for the world.

	(1) GRP	(2) Agriculture	(3) Manufacturing	(4) Services
RR	0.0000222	0.00000107	0.00000620	
TVAR	-0.0267	-0.00863	-0.00718	-0.0230
WD	-0.000303			-0.000237
SPEI	-0.0000903	-0.000281		-0.0000975
CW	-0.0000735		-0.000134	-0.000149
TX	-0.00223			-0.00282
HW		-0.000794		
RX		0.0000243		
TN		0.00637		
N	41799	32061	32147	32140
Adjusted R2	0.341	0.206	0.321	0.348
Adjusted within-R2	0.00325	0.00563	0.000626	0.00345

Notes: The table presents the results of the Post-LASSO regression. The dependent variable is the growth rate of gross regional product or of its sectoral equivalent. The independent variables are weather extremes and have been selected through Adaptive LASSO. All regressions include region fixed effects, year fixed effects, and regional linear time trends. Note that standard errors for the Post-LASSO estimates are invalid for inference.